

VayuVitals Cascading Multi-Horizon Atmospheric Predictive Engine

Mathematical Architecture, High-Order Atmospheric Physics, Adaptive Kalman Assimilation, and Empirical Validation on Delhi-NCR CPCB Continuous Network

1. Executive Summary & Model Identity

This specification documents the predictive machine learning engine integrated into the **VayuVitals** institutional health and clean air platform. Built upon continuous telemetry from the xKDR Forum CPCB regulatory monitoring network across Delhi-NCR (56,985 training observations and 14,247 held-out validation samples), the engine features a **Cascading Horizon Ensemble Ladder**, 24 high-order atmospheric physics features, **Log-Normal heteroscedastic target transformation**, and **Adaptive Kalman residual assimilation**. It provides empirical foresight for preemptive school closures, outdoor morning assembly restrictions, and institutional mitigation.

Model Name	vayuvitals-delhi-ncr-xgboost-v2
Engine Framework	Cascading Multi-Horizon Log-Normal XGBoost with Online Kalman Assimilation
S3 Artifact Path	s3://vayuvitals-aqi-dataset/models/model.tar.gz
Training / Test Data	56,985 Training Hours (80%) 14,247 Validation Hours (20%) — 71,232 Total Records
1h Rapid Nowcast Accuracy	21.05 µg/m³ MAE R² = 0.8664 (Pearson r = 0.9308) ±20 µg/m³: 67.2%
3h Morning Arrival Accuracy	29.29 µg/m³ MAE R² = 0.7936 (Pearson r = 0.8908) ±20 µg/m³: 50.9%

2. Mathematical Architecture: Log-Normal Target Transformation & Taylor Approximation

Atmospheric particulate concentrations exhibit strong positive skewness and multiplicative variance. To guarantee strictly positive predictions and stabilize heteroscedastic noise across seasons, the engine models the transformed variable $z = \ln(1 + \text{PM}_{2.5})$. The objective function minimizes regularized squared loss in log-space:

$$\text{Obj}(\theta) = \sum L(z_i, \hat{z}_i) + \sum [\gamma \cdot T_k + 0.5 \cdot \lambda \cdot \sum (w_j)^2 + \alpha \cdot \sum |w_j|]$$

where: $z_i = \ln(1 + y_i)$, $\hat{z}_i = \ln(1 + \hat{y}_i)$

- Taylor Expansion: $\text{Obj}^*(t) \approx \sum [g_i \cdot f_t(x_i) + 0.5 \cdot h_i \cdot (f_t(x_i))^2] + \Omega(f_t)$
- Optimal Leaf Weights: $w_j^* = -(\sum g_i) / (\sum h_i + \lambda)$
- Inverse Quantile Inversion:
 - P50 (Median Expected): $\hat{y}_{50} = \exp(\hat{z}) - 1$
 - P10 (Optimistic Dispersion): $\hat{y}_{10} = \max(15, \exp(\hat{z} - 1.28 \cdot \sigma_{\log}) - 1)$
 - P90 (Critical Inversion Surge): $\hat{y}_{90} = \exp(\hat{z} + 1.28 \cdot \sigma_{\log}) - 1$

Adaptive Kalman Filter / Online Residual Error Assimilation:

To assimilate real-time hyper-local sensor telemetry without boundary discontinuity, the forward engine computes the initial observation innovation $\text{residual}_0 = Y_{\text{sensor}}(0) - Y_{\text{synoptic}}(0)$ and exponentially relaxes it toward synoptic meteorological physics over an atmospheric decorrelation timescale ($\tau = 5.5$ hours):

$$\text{delta_residual}(t) = (Y_{\text{sensor}}(0) - Y_{\text{synoptic}}(0)) \cdot \exp(-t / \tau_{\text{decorrelation}})$$

where $\tau_{\text{decorrelation}} = 5.5$ hours (conforming to planetary boundary layer turnover in the Indo-Gangetic Plain).

3. Cascading Horizon Ensemble Performance Ladder

Empirical evaluation across 14,247 held-out validation hours demonstrates dramatic accuracy improvements for short lead times:

Forecast Lead Horizon	Target Operation	MAE	RMSE	Pearson r	R² Score	±20 µg/m³ Acc
-----------------------	------------------	-----	------	-----------	----------	---------------

t + 1h (Nowcast)	Emergency Recess & Bell Recall	21.05 µg/m ³	46.86 µg/m ³	0.9308	86.6%	67.2%
t + 3h (Arrival)	Morning School Bus & Transit	29.29 µg/m ³	49.93 µg/m ³	0.8908	79.4%	50.9%
t + 6h (Shift)	Institutional Handover & Sports	37.18 µg/m ³	55.23 µg/m ³	0.8332	69.4%	42.4%
t + 12h (Evening)	Evening Commute & Night Vent	43.54 µg/m ³	63.01 µg/m ³	0.7712	59.5%	37.3%
t + 24h (Day-Ahead)	Administrative Action & Closure	43.37 µg/m ³	64.19 µg/m ³	0.7625	58.2%	38.5%

4. High-Order Atmospheric Physics & Feature Vector (24 Dimensions)

The model ingests 24 engineered features capturing multi-scale atmospheric physics, chemical soot partitioning, and boundary layer dynamics:

Feature	Mathematical Formulation	Physics & Relative Importance
soot_mass	$PM_{2.5} \times (PM_{2.5} / PM_{10})$	26.4% Importance (#1): Isolates fine combustion carbon/soot from mineral dust.
doy_cos	$\cos(2\pi \cdot \text{DayOfYear} / 365.25)$	17.2% Importance (#2): Continuous macro-seasonal winter inversion depth.
pm25_now	Current observation ($\mu\text{g}/\text{m}^3$)	14.0% Importance (#3): High-frequency auto-regressive state anchor.
is_winter	Binary: Nov, Dec, Jan, late Oct	12.7% Importance (#4): Severe thermal radiation inversion boundary flag.
doy_sin	$\sin(2\pi \cdot \text{DayOfYear} / 365.25)$	5.7% Importance: Monsoon washout and transition phase tracking.
pm25_lag_1 / lag_2	$PM_{2.5}(t-1), PM_{2.5}(t-2)$	7.0% Combined: Short-term atmospheric memory and autoregression.
is_stubble_burning	Binary: Oct 20 - Nov 20	4.2% Importance: Regional agrarian biomass burning smoke window.
momentum_24h	$PM_{2.5}(t) - PM_{2.5}(t-24)$	3.0% Importance: 24-hour diurnal drift and synoptic building trend.
acceleration_2h	$(PM_{2.5}(t) - PM_{2.5}(t-1)) - (PM_{2.5}(t-1) - PM_{2.5}(t-2))$	2.3% Importance: Second-derivative plume onset acceleration rate.
u_wind, v_wind	$U = -S \cdot \sin(D), V = -S \cdot \cos(D)$	Vector wind decomposition tracking North-Westerly smoke advection.
inversion_intensity	$\max(0, (24 - T)/10) \times \text{stagnation_mult}$	Quantifies cold nighttime ground inversion pollutant trapping capacity.

5. Institutional Action & Civic Health Protocols

The model outputs feed directly into operational decision engines across Delhi-NCR schools, colleges, and healthcare centers:

- **6:30 AM Pre-Commute Advisory:** Delivers morning arrival risk, hourly confidence bands (P10–P90), and chemical source fingerprint to principals before student transit begins.
- **12:00 PM Mid-Day Emergency Alert:** Powered by Gemini AI, monitors sudden noon dust incursions and photochemistry to issue playground recall directives.
- **14-Day Chronic Non-Compliance Legal Action:** Aggregates empirical sensor data and forward forecasts into signed petitions addressed to the Directorate of Education.

Institutional Deployment Note: The VayuVitals engine operates with dual deployment parity: it is packaged as `model.tar.gz` in Amazon S3 for AWS SageMaker Serverless Inference and mirrored in the Node.js API server for resilient sub-millisecond local execution with continuous Kalman assimilation.